

SCRUM-27: MVP QA results

The frozen-v3 package records passing prototype checks for QA-01 through QA-21. All 36 fixture images were processed, but two images missed an expected odometer warning. Both affected rows still route to NEEDS REVIEW. Manual browser interaction and phone usability remain untested.

Scope and method

The MVP reader path and SCRUM-24 matrix were available. The supplied October 7 Jira export records SCRUM-9, SCRUM-24 and SCRUM-26 as Done. The review uses their synthetic images, fictional reference records and expected results. The 36-image set contains five valid forms and seven invalid logical cases, each in clean, rotated and perspective/shadow conditions. Eighteen images overlap the original SCRUM-9 pack.

Image-reading results are drawn from the recorded frozen-v3 QA run at revision 8680012. The October 7 review also included reruns of the 41 regression tests, 21 confidence-boundary cases, and 36 known-value cases. All 14 recorded reader, asset and configuration hashes match the repository revision reviewed below.

Functional results

Check group	Result	Evidence and interpretation
QA-01 through QA-21	Passed	Recorded matrix checks for labels, orientation, data controls, extraction, assembly, validation and routing.
Component / regression tests	41 / 41 passed	Covers crop uncertainty, preprocessing warnings, alignment and reviewed corrections.
Confidence boundaries	21 / 21 passed	Checks below, equal to and above the ordinary and critical odometer thresholds.
Known-value fixture checks	36 / 36 passed	Supplies written answers at confidence 0.99. Expected flags and routes match.
Image-path processing	36 / 36 completed	Recorded run registers all images and extracts all 5,328 expected cells, with no execution errors.
Expected source-fault detection	19 passed; 2 failed	Of 21 invalid images, two miss one intended warning. See the odometer finding on page 2.

These groups overlap and use different units; their counts are not added into one pass total. Known-value checks do not measure recognition. QA-22 is Evaluated, with no assumed accuracy pass threshold.

Version and execution conditions

Frozen reader implementation	5e95a2cbdbe7d86d32d3aa9d88d3db77d7a9db2c
Recorded source / fixture snapshot	208a29981c65ce8121302a5fc05ee9e326aa2f44
Frozen QA evidence revision	868001264d79b43646a433a728214b81f0205e1f
Repository reviewed October 7	0b7096a07d1a1c0479401539fd965bf0ea9fac8d

Frozen model SHA-256: 9da5b69c57476cae69a266553c3b1d39c35cd0ae1306716ed586270a92d3dfa9

Recorded frozen run: October 7, 2026; business-validation date October 5, 2026; Python 3.14.6, NumPy 2.5.3, OpenCV 4.14.0 and Matplotlib 3.11.2. Review reruns: Python 3.12.14, NumPy 2.3.5, OpenCV 4.13.0 on Linux. Frozen confidence settings: 0.82 ordinarily and 0.92 for each odometer's first three digits.

Findings and recorded measurements

Odometer warning missed in two images

In `invalid_odometer_clean` and `invalid_odometer_rotated`, row 1 contains start 068253, end 068252 and miles 037. The reader returns start 062253, so `odometer_end_after_start` passes on the misread values. The miles-difference check still fails and the affected row routes to NEEDS REVIEW. Known-value validation detects the intended faults. These remain two failed image checks caused by a shared recognition limitation.

Review or reproduce the cases using the procedure on page 3. Compare the truth and prediction for row 1 in [invalid_odometer_clean.json](#) and [invalid_odometer_rotated.json](#). The expected and observed failed-check lists are in [QA_Fixture_Metrics.json](#).

Routing outcome

The 36-image frozen run produced 26 AUTO-POST candidates and 190 NEEDS REVIEW rows. No invalid-source rows, extra predicted rows, or header/week/footer/trip-row transcription errors were accepted among those 26 candidates. Routing applies to individual rows: an unaffected valid row in a form containing a different faulty row can remain eligible. These small, related samples do not establish calibrated confidence or production payment safety.

Model measurements kept separate from functional checks

Synthetic dataset	Correct characters	Exact fields	Exact trip rows
SCRUM-9: 18 images	2,590 / 2,664 (97.22%)	526 / 594	60 / 108
SCRUM-26: 36 images	5,222 / 5,328 (98.01%)	1,089 / 1,188	139 / 216
Shared valid subset: 15 images	2,152 / 2,220 (96.94%)	433 / 495	47 / 90

The table reports frozen-v3 recognition measurements separately from functional test outcomes. The 18-image pack has 74 character mismatches; the 36-image pack has 106. Both report no missing expected cells. Thus successful extraction does not establish correct recognition. The shared valid subset contains five underlying forms, each in three conditions. Overlapping images and related variants are not independent samples; the higher 36-image score reflects the case mix, not an improved reader. SCRUM-18 model benchmarking remains separate.

Approved classroom scope

The October 7 SCRUM-22 record contains the Product Lead's approval to retain the frozen conservative policy, the QA acceptance comment, and Done status. Independent calibration remains follow-up work before production use. AUTO-POST means a prototype candidate; the application does not issue payments or post to accounting.

Untested scope and reproduction

Not performed in this review

Scope	Status and reason
Manual browser interaction	Not performed. Upload/preview/button behavior and correction usability were not exercised in a working browser session. The web interface is for demonstrations; the command-line interface is the default/fallback. These additional interface checks are deferred for this report.
Phone layout and camera use	Not performed. No physical-phone session verified readability, touch controls, the file picker, or taking and submitting a new photo. Existing saved-image results do not establish those behaviors. These additional demonstration checks are deferred with the browser checks.
Independent calibration / new writers	Deferred under the approved classroom scope. This requires new labeled forms and separate calibration and evaluation data before production use. No fresh handwriting-generalization claim is made.

Full data audits, visual QA and isolated-character evaluation use the recorded package evidence. Untested scope is not counted as passed.

Review procedure and reproduction

Evidence review: compare the execution record with the frozen reader, checkpoint and policy identities. Inspect QA-01 through QA-21 outcomes, the 21 confidence-boundary inputs and expected routes, the 36 known-value expected flags/routes, and each image case's source-fault and candidate checks. Validation checks used the recorded controlled inputs, written fixture values, policy, reference records and October 5 business date. The regression command was:

```
python -m unittest discover -s tests -v
```

For a new full frozen QA run, use a separate clone at the evidence revision below. Follow the repository setup, restore the checkpoint matching the SHA-256 on page 1, and use the recorded package versions and five EMNIST checksums. Then run:

```
git checkout 868001264d79b43646a433a728214b81f0205e1f
.venv/bin/python scripts/run_qa.py --output-dir outputs/qa_v3_reproduction
```

On Windows, use .venv\Scripts\python.exe. The runner verifies the frozen inputs and executes notebook Steps 3-22 in an isolated copy without training. Colab-only setup and archive-download cells are skipped locally; this is not skipped functional coverage. Inspect QA-02/QA-04 visual outputs.

Evidence references

[Frozen QA package and matrix](#) | [Execution record and source hashes](#) | [Recorded environment](#)
[SCRUM-9 image measurements](#) | [SCRUM-26 image measurements](#) | [Known-value checks](#)
[Confidence-boundary evidence](#) | [Regression log](#) | [SCRUM-22 scope and acceptance](#)